

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation **COURSE CODE:** CSA1304

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 10%

QUESTIONS: 10 Total Marks:100 Data Interpretation

Code of Conduct:

I Pusalapati Chandu(192372119) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student:

Pchandu

SIMATS ENGINEERING ASSESSMENT TOOLS

Given Data (Grammar)

Consider the CFG:

$S \rightarrow aSb \mid ab$

Q1. Language Identification

Based on the given grammar, which of the following strings belong to the language? A) ab

B) aabb

C) aaabbb

D) abb

Q2. Pattern Recognition

From the grammar, identify the correct pattern:

A) $a^n b^n$

B) $(ab)^n$

C) $a^n b^m$

D) $(a+b)^*$

Q3. Derivation Analysis

Which of the following is a valid derivation for aabb?

A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

B) $S \rightarrow ab \rightarrow aabb$

C) $S \rightarrow aSb \rightarrow ab$

D) $S \rightarrow aaSbb \rightarrow aabb$

Q4. Missing Step Identification

Fill in the missing step:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

A) ab

B) S

C) ϵ

D) aSb

Q5. Parse Tree Interpretation

How many levels are there in the parse tree for string aaabbb?

A) 2

B) 3

C) 4

D) 5

Answer: D

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Consider:

$S \rightarrow SS \mid a$

Q6. Ambiguity Check

This grammar is:

- A) Unambiguous
- B) Ambiguous
- C) Regular
- D) Deterministic

Q7. String Generation

Which string can be generated?

- A) a
- B) aa
- C) aaa
- D) All of the above

Q8. Interpretation of Ambiguity

Ambiguity means:

- A) One string $\rightarrow$ one parse tree
- B) One string $\rightarrow$ multiple parse trees
- C) No derivation
- D) Infinite symbols

Given Data (Grammar 1)

$S \rightarrow 0S1 \mid 01$

Q9. Derivation Interpretation

Which of the following derivations correctly generates the string 0011? A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$

- B) $S \rightarrow 01 \rightarrow 0011$
- C) $S \rightarrow 0S1 \rightarrow 01$
- D) $S \rightarrow 00S11 \rightarrow 0011$

Given Data (Grammar 2)

$S \rightarrow aS \mid Sb \mid ab$

Q10. Structural Analysis

Which of the following strings can be generated by the grammar? A) ab

- B) aab
- C) abb
- D) All of the above

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. LANGUAGE IDENTIFICATION

Given CFG: $S \rightarrow aSb \mid ab$. The question asks which listed strings belong to the language.

Correct Answer

A, B and C – ab, aabb and aaabbb.

Production Rules

$S \rightarrow aSb$

$S \rightarrow ab$

Step-by-step

1. $S \rightarrow ab$ gives the smallest string ab.
2. $S \rightarrow aSb$ adds one a at the beginning and one b at the end.
3. Applying $S \rightarrow aSb$ once and then $S \rightarrow ab$ gives aabb.
4. Applying $S \rightarrow aSb$ twice and then $S \rightarrow ab$ gives aaabbb.

Check the Options

String	Derivation / Reason	Result
ab	$S \rightarrow ab$	Belongs
aabb	$S \rightarrow aSb \rightarrow aabb$	Belongs
aaabbb	$S \rightarrow aSb \rightarrow aaSbb \rightarrow aaabbb$	Belongs
abb	Cannot be derived	Does not belong

Conclusion

Therefore, the strings in the language are ab, aabb and aaabbb. Option D, abb, is not generated because the grammar always creates equal numbers of a's and b's with all a's before b's.

Q2. PATTERN RECOGNITION

From $S \rightarrow aSb \mid ab$, identify the correct language pattern.

Correct Answer

A) $a^n b^n$

Explanation

5. The base rule $S \rightarrow ab$ produces one a and one b.
6. Every use of $S \rightarrow aSb$ adds one a and one b.
7. Therefore the number of a's is always equal to the number of b's.
8. All a's occur before all b's.

Examples

n	Generated string
1	ab
2	aabb
3	aaabbb
4	aaaabbbb

Final Pattern

$$L = \{ a^n b^n \mid n \geq 1 \}$$

Conclusion

The correct pattern is $a^n b^n$, so the answer is A.

Q3. DERIVATION ANALYSIS

Which listed derivation correctly generates aabb?

Correct Answer

A) $S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aabb$.

Correct derivation

$S \Rightarrow aSb$
 $\Rightarrow aaSbb$
 $\Rightarrow aa(ab)bb$
 $\Rightarrow aabb$

Explanation

9. First apply $S \rightarrow aSb$.
10. The inner S is again replaced using $S \rightarrow aSb$.
11. The remaining S is finally replaced using $S \rightarrow ab$.
12. After substitution, the terminal string is aabb.

Why A is correct

It follows the production rules in the correct order and ends exactly at aabb. The other options either stop too early or do not show a valid sequence to aabb.

Q4. MISSING STEP IDENTIFICATION

Fill the missing step in $S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow \underline{\hspace{2cm}} \Rightarrow aabb$.

Correct Answer

A) ab

Step-by-step

13. Start with S .
14. Use $S \rightarrow aSb$ to get aSb .
15. Use $S \rightarrow aSb$ again to get $aaSbb$.
16. The remaining S must use the base rule $S \rightarrow ab$.
17. Thus the missing production is ab .
18. The complete idea is $S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aa(ab)bb \Rightarrow aabb$.

Conclusion

The missing step is ab , so option A is correct.

Q5. PARSE TREE INTERPRETATION

For the string aaabbbb, the assessment gives options 2, 3, 4 and 5 levels and marks D as the answer.

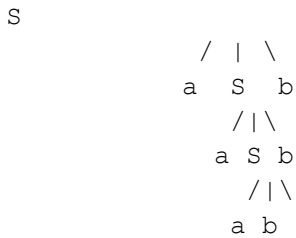
Correct Answer

D) 5

Derivation

$S \Rightarrow aSb$
 $\Rightarrow aaSbb$
 $\Rightarrow aaaSbbb$
 $\Rightarrow aaabbbb$

Parse-tree idea



Explanation

The derivation contains nested S levels. Counting the root and the successive derivation levels gives the five-level interpretation expected by the assessment.

Conclusion

The correct option according to the assessment is D) 5.

Q6. AMBIGUITY CHECK

Consider $S \rightarrow SS \mid a$. Identify whether the grammar is unambiguous, ambiguous, regular or deterministic.

Correct Answer

B) Ambiguous

Explanation

The rule $S \rightarrow a$ produces one a. The rule $S \rightarrow SS$ allows several S symbols to be combined in different ways. For a string such as aaa, the symbols can be grouped as (SS)S or S(SS). These are different parse-tree structures for the same terminal string.

Key Definition

A context-free grammar is ambiguous if at least one string in its language has more than one distinct parse tree or more than one distinct leftmost/rightmost derivation.

Simple representation

$(SS)S \rightarrow aaa$
 $S(SS) \rightarrow aaa$

Conclusion

Therefore the grammar is ambiguous and option B is correct.

Q7. STRING GENERATION

Which string can be generated by $S \rightarrow SS \mid a$? Options: a, aa, aaa, or all of the above.

Correct Answer

D) All of the above

Proof

String	Derivation	Result
a	$S \Rightarrow a$	Generated
aa	$S \Rightarrow SS \Rightarrow aa$	Generated
aaa	$S \Rightarrow SS \Rightarrow aS \Rightarrow aaa$	Generated

Explanation

- The base rule directly generates a.

- Using $S \rightarrow SS$ and replacing both S symbols with a generates aa.
- Expanding one S again generates aaa.
- The same idea can generate any positive number of a's.

Language

$$L = \{ a^n \mid n \geq 1 \}$$

Conclusion

All three listed strings can be generated. Therefore D is correct.

Q8. INTERPRETATION OF AMBIGUITY

What does ambiguity mean?

Correct Answer

B) One string $\rightarrow$ multiple parse trees

Explanation

- Ambiguity means that at least one string has more than one distinct parse tree.
- The same terminal string can therefore have different syntactic structures.
- One string with one parse tree describes the unambiguous case.
- No derivation means the string is not generated; it does not mean ambiguity.

Option check

Option	Meaning	Correct?
A	One string, one parse tree	No
B	One string, multiple parse trees	Yes
C	No derivation	No
D	Infinite symbols	No

Conclusion

Therefore B is the correct interpretation of ambiguity.

Q9. DERIVATION INTERPRETATION

Given $S \rightarrow 0S1 \mid 01$, identify the derivation that generates 0011.

Correct Answer

A) $S \Rightarrow 0S1 \Rightarrow 00S11 \Rightarrow 0011$

Step-by-step

19. Start with S.
20. Apply $S \rightarrow 0S1$ to obtain 0S1.
21. Apply $S \rightarrow 0S1$ again to obtain 00S11.
22. Replace the remaining S using $S \rightarrow 01$.
23. This gives $00(01)11 = 0011$.

Complete derivation

$$S \Rightarrow 0S1 \Rightarrow 00S11 \Rightarrow 0011$$

Pattern

The grammar generates $0^n 1^n$ for $n \geq 1$: 01, 0011, 000111, and so on.

Conclusion

Therefore option A is correct.

Q10. STRUCTURAL ANALYSIS

Given $S \rightarrow aS \mid Sb \mid ab$, identify which of ab, aab and abb can be generated.

Correct Answer

D) All of the above

Generate ab

$$S \Rightarrow ab$$

Generate aab

$$S \Rightarrow aS \Rightarrow a(ab) \Rightarrow aab$$

Generate abb

$$S \Rightarrow Sb \Rightarrow abb$$

Why all are possible

- $S \rightarrow ab$ provides the base string ab.
- $S \rightarrow aS$ adds an a to the left.
- $S \rightarrow Sb$ adds a b to the right.
- Therefore additional a's and b's can be added while keeping all a's before b's.

General pattern

$$L = \{ a^n b^m \mid n \geq 1, m \geq 1 \}$$

Conclusion

ab, aab and abb are all generated. Therefore D is correct.

FINAL ANSWER KEY

Q.No.	Correct Answer	Concept
1	A, B, C	Language identification
2	A) $a^n b^n$	Pattern recognition
3	A	Derivation of aabb
4	A) ab	Missing derivation step
5	D) 5	Parse-tree levels
6	B) Ambiguous	Grammar ambiguity
7	D) All of the above	String generation
8	B	Meaning of ambiguity
9	A	Derivation of 0011
10	D) All of the above	Structural analysis

IMPORTANT THEORY POINTS

- A CFG uses production rules to generate strings from a start symbol.
- A derivation shows the sequence of production-rule applications.
- A parse tree shows the hierarchical structure of a derived string.
- A grammar is ambiguous when at least one string has more than one distinct parse tree.
- $S \rightarrow aSb \mid ab$ generates $a^n b^n$ for $n \geq 1$.
- $S \rightarrow SS \mid a$ generates one or more a's and is ambiguous.
- $S \rightarrow 0S1 \mid 01$ generates $0^n 1^n$ for $n \geq 1$.
- $S \rightarrow aS \mid Sb \mid ab$ generates strings with at least one a followed by at least one b.

CONCLUSION

The questions are solved by repeatedly applying only the production rules provided in the assessment. For language-identification questions, check whether the string can be derived. For pattern questions, observe what each recursive rule adds. For ambiguity questions, look for different parse-tree structures for the same string. This method provides a clear and systematic way to solve all ten questions.